

PHASE 3: INPUT, OUTPUT AND REPORT DESIGN

For Yisakal SACCO Management System

This chapter explains the screens and reports used in the Yisakal SACCO system, such as forms, menus, and dashboards.

Its main goal is to make sure that everything users see and interact with on the screen directly matches the database structure behind the system. This includes input fields, selections, tables, and reports.

In short, it ensures that what users enter and view in the system is properly connected to the organized database so that data is stored correctly, consistently, and without errors.

1. Input Forms Design (Data Accepting Interfaces)

This part explains how new member information is entered into the SACCO system using a registration form.

The form is used by staff to officially add a new client into the system. When the form is submitted, the system safely saves the member's details into the database in one complete step and makes sure there are no duplicates (for example, using a unique ID like a national identity number).

YISAKAL SACCO — NEW MEMBER REGISTRATION PORTAL

PERSONAL INFORMATION

First Name: Abebe
Middle Name: Kebede
Last Name: Balcha
Gender:
☒ Male
☐ Female
Date of Birth: 1988-04-12
Credit Score: 720

IDENTIFICATION & CONTACTS

National ID #: NID-8830129-ETH
Kebele ID #: KBL-04-99302
Phone Number: 0911234567
Email Address: abebe.kebede@gmail.com
Occupation: Senior Agricultural Procurement Consultant

RESIDENTIAL ADDRESS (COMPOSITE UNPACKING)

Region: Addis Ababa
City: Addis Ababa
Subcity: Kirkos
Kebele: 04
House #: 921/B

SYSTEM PARAMETERS

Home Branch: Addis Ababa Central Branch
Membership Date: 2026-05-22
Account Status:
☒ Active
☐ Inactive / Suspended

CLEAR FIELDS
SUBMIT REGISTRATION APPLICATION

This section explains a form used to create a savings account for an existing member.

It is used when a member wants to open a savings account that earns interest. The form connects the account to the correct savings product type, links it to the right member, and records where and when the account was opened.

Form 2: Savings Account Guided Setup Tool (Provision Wizard)

This section explains a form used to create a savings account for an existing member.

It is used when a member wants to open a savings account that earns interest. The form connects the account to the correct savings product type, links it to the right member, and records where and when the account was opened.

YISAKAL SACCO — SAVINGS ACCOUNT PROVISIONING WIZARD

MEMBER CROSS-REFERENCE LOOKUP

Search Member ID:
M-10029384
VALIDATE MEMBER

Verified Legal Name:
Abebe Kebede Balcha [ACTIVE]

PRODUCT CONFIGURATION & FINANCIAL BASELINING

Savings Product Tier:
Fixed Term High-Yield Deposit (12-Month)

Assigned Branch:
Addis Ababa Central Branch (BR-01)

System Account Code:
ACC-10029384

Initial Opening Ledger:
5,000.00 ETB (Minimum balance required: 5,000.00 ETB)

Opening Timestamp:
2026-05-22

Initial Account State:
☒ Active (Authorized Post-Deposit)
☐ Dormant / Locked

CANCEL WIZARD
PROVISION ACCOUNT LEDGER

Form 3: Credit Application and Collateral Intake Dashboard

This form is used when a member applies for a loan and provides security for it.

It helps the system record the loan request, link it to any assets used as collateral, and include any guarantors who agree to take responsibility if the borrower fails to repay.

YISAKAL SACCO — CREDIT APPLICATION & COLLATERAL PROCESSING CENTRE

BORROWER PROFILE LINKAGE

Primary Borrower ID:
M-88392

Borrower Full Name:
Chernet Assefa

Target Credit Product:
Capital Growth Business Loan

Base Interest Matrix:
12.5% per Annum (Reducing Balance)

CORE LOAN UNDERWRITING METRICS

Requested Principal:
250,000.00

Currency Unit:
Ethiopian Birr (ETB)

Repayment Term:
24

Time Unit Anchor:
Calendar Months

Disbursement Date:
2026-06-01

Calculated Maturity:
2028-06-01

COLLATERAL SECURITY ASSETS ASSET REGISTRY SEGMENT

Asset Category:
Automobile / Vehicle Title Deed

Estimated Market Value:
350,000.00

Ownership Doc Ref #:
DEF-993021-2026-AA

Comprehensive Specs:
Toyota Vitz 2015 Model, Motor Capacity: 1300cc, Plate No: AA-3-A12345, Chassis: KMVH42BA18, Excellent Condition. Certified by Federal Transport Authority.

GUARANTOR LIABILITY UNDERWRITE ENDORSEMENT GRID

Guarantor Member ID:
M-10293

Guarantor Legal Name:
Martha Girma

Committed Underwrite:
100,000.00

Workflow Verification State:
Pending Signature Verification

VOID APPLICATION

RUN CREDIT RISK ENGINE & COMPILE

2. Output Forms Design (Transaction views and Data Storing(in the DB))

This section shows how the database actually store data in the database system after accepting the input forms.

1.Branch Table

	branch_id	name	region	city	subcity	kebele	street	phone_number	manager_id	created_date	status
▶	1	Merkato Branch	Addis Ababa	Addis Ababa	Addis Ketema	05	Merkato Ave	0111234567	NULL	2020-01-15	Active
	2	Bole Branch	Addis Ababa	Addis Ababa	Bole	03	Bole Rd	0119876543	NULL	2020-06-01	Active
	3	Bahir Dar Branch	Amhara	Bahir Dar	Fasilo	01	Lake Tana St	0581234567	NULL	2021-03-10	Active
	4	Jimma Branch	Oromia	Jimma	Jimma	02	Main St	0471112233	NULL	2021-08-12	Active
	5	Hawassa Branch	SNNPR	Hawassa	Tabor	04	Unity St	0462223344	NULL	2022-02-10	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2.Employee Table

	employee_id	branch_id	full_name	email	phone_number	role	hire_date	status
▶	101	1	Abebe Kebede	abebe@yisakal.com	0911111111	Manager	2020-01-20	Active
	102	1	Sara Tesfaye	sara@yisakal.com	0922222222	Officer	2020-03-15	Active
	103	2	Dawit Haile	dawit@yisakal.com	0933333333	Teller	2021-01-10	Active
	104	2	Meron Alemu	meron@yisakal.com	0944444444	Manager	2020-06-05	Active
	105	3	Yonas Girma	yonas@yisakal.com	0955555555	Officer	2021-04-01	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.SavingAccountProduct Table

	product_id	name	description	min_balance	interest_rate	posting_frequency	status
▶	301	Regular Savings	Standard account	100.00	7.00	Yearly	Active
	302	Youth Savings	Youth account	50.00	8.50	Monthly	Active
	303	Fixed Deposit	12 months deposit	5000.00	10.00	Yearly	Active
	304	Business Savings	Business account	1000.00	9.00	Monthly	Active
	305	Women Savings	Women empowerment account	200.00	8.00	Monthly	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4.LoanProduct Table

	product_id	name	description	min_amount	max_amount	min_tenure_months	max_tenure_months	base_interest_rate	interest_type	requires_guarantor	requires_collateral	status
▶	401	Personal Loan	Personal needs	5000.00	100000.00	6	36	15.00	Reducing	1	0	Active
	402	Business Loan	Business support	50000.00	500000.00	12	60	12.50	Flat	1	1	Active
	403	Emergency Loan	Short term	1000.00	20000.00	1	6	18.00	Flat	0	0	Active
	404	Education Loan	School fees	2000.00	50000.00	6	24	10.00	Reducing	1	0	Active
	405	Agriculture Loan	Farm support	3000.00	150000.00	6	48	11.00	Reducing	1	1	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5.Member Table

	member	full_name	branch_id	gender	date_of_birth	national_id_no	kebele_id_number	email	phone_number	region	city	subcity	kebele	street	house_number	occupation	credit	membership	status
▶	205	Fatun...	3	Female	1995-12-30	NID1005	KID1005	fatuma@gm...	0923456789	Amhara	Bahir Dar	Fasilo	01	Univer...	15	Acco...	710	2022-0...	Active
	204	Solom...	2	Male	1988-02-14	NID1004	KID1004	solomon@g...	0912345678	Addis A...	Addis ...	Yeka	10	Megen...	88	Engin...	790	2022-0...	Active
	203	Helen ...	2	Female	1992-08-25	NID1003	KID1003	helen@gmail...	0988888888	Addis A...	Addis ...	Bole	07	Camer...	33	Nurse	750	2021-0...	Active
	202	Kebed...	1	Male	1985-11-03	NID1002	KID1002	kebede@gm...	0977777777	Addis A...	Addis ...	Arada	02	Piassa ...	5	Teacher	680	2021-0...	Active
	201	Tigist ...	1	Female	1990-05-12	NID1001	KID1001	tgist@gmail...	0966666666	Addis A...	Addis ...	Addi...	05	Merkat...	12A	Trader	720	2021-0...	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

6.SavingAccount Table

	account_id	member_id	account_number	branch_id	product_id	current_balance	open_date	status
▶	501	201	SA1001	1	301	25000.00	2021-01-10	Active
	502	202	SA1002	1	301	12500.00	2021-02-20	Active
	503	203	SA1003	2	302	8000.00	2021-06-15	Active
	504	204	SA1004	2	303	50000.00	2022-01-05	Active
	505	205	SA1005	3	304	15000.00	2022-03-20	Active
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

7.Loan Table

	loan_id	member_id	product_id	loan_officer_id	principal_amount	interest_rate	term_months	disbursement_date	maturity_date	outstanding_principal	outstanding_interest	outstanding_fees	status
▶	601	201	401	102	50000.00	15.00	24	2023-01-15	2025-01-15	30000.00	4500.00	0.00	Active
	602	203	403	103	10000.00	18.00	3	2024-06-01	2024-09-01	0.00	0.00	0.00	Closed
	603	204	402	104	200000.00	12.50	36	2024-03-10	2027-03-10	170000.00	21250.00	500.00	Active
	604	205	401	105	30000.00	15.00	12	NULL	NULL	30000.00	0.00	0.00	Approved
	605	202	404	101	20000.00	10.00	18	2024-01-10	2025-07-10	12000.00	1800.00	0.00	Active
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

8.LoanSchedule Table

	schedule_id	loan_id	installment_number	due_date	principal_due	interest_due	fees_due	principal_paid	interest_paid	fees_paid	status
▶	701	601	1	2023-02-15	2083.00	625.00	0.00	2083.00	625.00	0.00	Paid
	702	601	2	2023-03-15	2083.00	598.00	0.00	2083.00	598.00	0.00	Paid
	703	601	3	2023-04-15	2083.00	572.00	0.00	0.00	0.00	0.00	Overdue
	704	603	1	2024-04-10	5555.00	2083.00	0.00	5555.00	2083.00	0.00	Paid
	705	605	1	2024-02-10	1111.00	166.00	0.00	1111.00	166.00	0.00	Paid

9.SavingTransaction Table

	transaction_id	account_id	transaction_type	amount	transaction_date	balance_after_transaction	reference_number	employee_id
▶	801	501	Deposit	10000.00	2021-01-10 09:00:00	10000.00	STX1	103
	802	501	Deposit	15000.00	2021-06-15 10:30:00	25000.00	STX2	103
	803	502	Deposit	20000.00	2021-02-20 11:00:00	20000.00	STX3	103
	804	502	Withdrawal	7500.00	2022-08-10 14:15:00	12500.00	STX4	103
	805	503	Deposit	8000.00	2021-06-15 09:45:00	8000.00	STX5	103

10.LoanTransaction Table

	transaction_id	loan_id	loan_schedule_id	transaction_type	amount	transaction_date	payment_method	reference_number	employee_id	reversal_status
▶	901	601	NULL	Disbursement	50000.00	2023-01-15 08:00:00	Transfer	LTX1	102	None
	902	601	701	Repayment	2708.00	2023-02-15 09:30:00	Cash	LTX2	103	None
	903	603	NULL	Disbursement	200000.00	2024-03-10 08:30:00	Transfer	LTX3	104	None
	904	603	704	Repayment	7638.00	2024-04-10 11:00:00	Transfer	LTX4	103	None
	905	605	NULL	Disbursement	20000.00	2024-01-10 10:00:00	Transfer	LTX5	101	None

11.Guaranty Table

	loan_id	member_id	guarantee_amount	status
▶	601	202	25000.00	Active
	603	203	100000.00	Active
	604	201	15000.00	Active
	605	204	10000.00	Active
•	NULL	NULL	NULL	NULL

12.Collateral Table

	collateral_id	loan_id	collateral_type	description	estimated_value	ownership_document_ref	status
▶	1101	603	Vehicle	Toyota Vitz	350000.00	DOC1	Pledged
	1102	603	Property	Land 200sqm	500000.00	DOC2	Pledged
	1103	605	House	Small house	300000.00	DOC3	Pledged
	1104	602	Equipment	Office equipment	50000.00	DOC4	Released
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

13.Audit Table

	audit_id	entity_name	entity_id	action_type	old_values	new_values	employee_id	created_at	ip_address
▶	1201	Loan	601	Create	NULL	{"status": "Applied"}	102	2023-01-15 08:00:00	192.168.1.10
	1202	Loan	601	Update	{"status": "Applied"}	{"status": "Disbursed"}	101	2023-01-15 08:05:00	192.168.1.11
	1203	Member	201	Create	NULL	{"status": "Created"}	102	2021-01-10 09:00:00	192.168.1.10
	1204	Loan	603	Update	{"status": "Approved"}	{"status": "Active"}	104	2024-03-10 09:00:00	192.168.1.12
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

14.FeeType Table

	fee_type_id	name	description	calculation_method	amount_or_rate	is_active
▶	1301	Processing Fee	Loan fee	Percentage	2.00	1
	1302	Late Fee	Penalty	Percentage	5.00	1
	1303	Service Fee	Monthly fee	Flat	25.00	1
	1304	Registration Fee	Joining fee	Flat	100.00	1
•	NULL	NULL	NULL	NULL	NULL	NULL

15.FeeEvent Table

	fee_event_id	fee_type_id	loan_id	saving_transaction_id	paid
▶	1401	1301	601	NULL	1
	1402	1302	601	NULL	0
	1403	1301	603	NULL	1
	1404	1304	605	NULL	1
•	NULL	NULL	NULL	NULL	NULL

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16.FeeTransaction Table

	fee_transaction_id	fee_event_id	amount	reference	transaction_date
▶	1501	1401	1000.00	FTX1	2023-01-15 08:10:00
	1502	1403	4000.00	FTX2	2024-03-10 08:40:00
	1503	1404	500.00	FTX3	2024-01-10 08:20:00
★	NULL	NULL	NULL	NULL	NULL

3. Management and Reporting Statements

Analytical reports process historical atomic data rows into aggregated summaries designed to assist management audits, monitoring of risk vectors, and regulatory reporting.

Report : Amortization Schedule Ledger Matrix

Purpose: A document showing the payment timeline and principal reduction for a borrower over the life of a loan. This structural matrix maps directly to rows inside the LoanSchedule weak entity table, which are automatically generated by the credit engine once a loan application transitions to an Active state.

YISAKAL COOPERATIVE CREDIT & SAVING SOCIETY LTD.

Structured Amortization & Repayment Timeline Matrix

Loan Reference Code:	LN-2026-00492	Total Issued Value:	250,000.00 ETB
Borrowing Entity ID:	M-88392 (Chernet Assefa)	Interest Rate Engine:	12.5% per Annum (Reducing Balance)
Maturity Threshold:	24 Months (24 Installments)	Generation Account Date:	2026-05-22 01:53 AM EAT

Inst #	Due Date	Principal Due (ETB)	Interest Due (ETB)	Fees Due	Total Due (ETB)	Lifecycle State
01	2026-07-01	10,416.67	2,604.17	150.00	13,170.84	UNPAID
02	2026-08-01	10,416.67	2,495.66	0.00	12,912.33	UNPAID
03	2026-09-01	10,416.67	2,387.15	0.00	12,803.82	UNPAID
...	...	...	...	...	...	...
24	2028-06-01	10,416.59	108.51	0.00	10,525.10	UNPAID
AGGREGATE TOTALS:		250,000.00	32,552.12	150.00	282,702.12	

4. Conclusion

This chapter explains how the system screens, forms, receipts, and reports are connected to the database structure. It ensures that everything users enter, view, or print in the Yisakal SACCO system is correctly linked to the right database information and follows the rules designed for the system.

In general, it means that the system interface and the database are carefully connected so the SACCO operations work accurately, consistently, and reliably.